1/1/24 1) Void-main()
Ramin Sir

```
{
  static int i=8;
  if (--i){
    main();
    pf("%d", i); }}
```

A:) 0
0
0
0

data
| i=8 |
1000

if i==0
A:) 0
0
0
0

2) The given expression used as int (*a)[8] ?

A:) varying row & fixed column.

int a[4][10];
fixed row & fixed column
int a[ ][7];
varying row & fixed column

3) Nested unions & self referential unions are not possible in C.

A:) False.

4) The con

5) fprintf, fread, fwrite.

6) fseek.

7) what is the difference between the
include <file.h> ⇒ predefined data type
include "file.h" ⇒ userdefined data type.

:)
```
main(){
char a[5][5], flag;
  a[0][0] = 'A';
  flag = ((a==*a) && (*a==a[0]));
  pf ("%d", flag); // 1
}
```

A:) a = 1000
a[0] = 1000
*(a+0) ⇒ a[0] ⇒ 1000

flag = 1

enum ← integer

) cross compiler.

scanf() ⇒ error

```
int scanf =500;
pf ("%d\n", scanf); //500.
```

x=111, y=222

```
if(!(!x) && y)
  pf ("x=%d\n", x); // 111 ⇒ Ans.
else
  pf("y=%d\n", y);
```

```
int my-fun();
main()
{
  int *ptr = my_fun();
  pf("%d\n", *ptr); //segmentation fault.
}
int my_fun()
{
  int num=10;
  return num
}
```

| 10 |
ptr 1000

| 10 |
num 2000

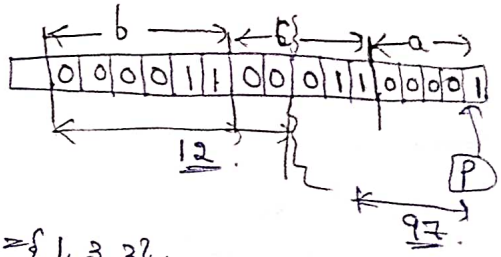
p=10

14)
```
int EOF = 11;
pf("%d\n", EOF); //error
```

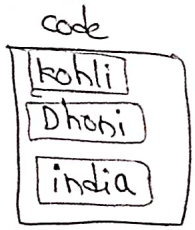
15)
```
struct Bangaluru
{
    int RCB;
};
main()
{
    struct Bangaluru star;
    star.RCB = -12;
    pf("%d\n", star.RCB>>3); // -2
}
```

-12 >> 3

16)
```
main(){
struct bitfield {
    unsigned a : 5;
    unsigned c : 5;
    unsigned b : 6;
} bit;
char *p;
struct bitfield *ptr, bitI = {1, 3, 3};
p = &bitI;
++p;
pf("%d", *p); //12
--p;
pf("%d\n", *p); //97
}
```



17)
```
struct student
{
    char * name;
};struct student s;
struct student w-cup(void)
{
    s.name = "kohli";
    pf("%s", s.name);
    s.name = "dhoni";
    return s;
}
main()
{
    struct student m = w_cup();
    pf("%s", m.name);
    m.name = "india";
}   pf("%s\n", s.name);
```



// kohli dhoni dhoni

18)
```
main()
{
  struct ABC
  {
    int abc:2;
  } kk;
  kk.abc=3;
  Pf("%d\n", kk.abc); // -1.
}
```

sign 11 0 1 → -1 ⟹ -1

19)
```
struct leader
{
  char *lead;
  int born;
};
struct leader_15 = {"Abdulkalam", 1931};
struct leader_oct = _15;
Pf("%s %d\n", _oct.lead, _15.born);
}
```

A:-) Abdulkalam, 1931

20)
```
main()
{
  char ch=0;
  while (ch=='0')
  {
    Pf("Loop");
    break;
  }
  Pf("LAB\n"); // LAB
}
```

21)
```
int main()
{
  int int_=11;
  for( ; ; int_++)
  {
    Pf("%d\n", int_);
    int_=10;
    if (int_>=10)
      break;
  }
}
```

ANS:-) 11.

22)
```
main()
{
  char array[6]="pawan", *ptr, *ptr1;
  ptr=array;
  ptr1=ptr+3;
  *ptr1=97;
  Pf("ptr=%s, ptr1=%s\n", ptr, --ptr1); // pawan, wan
}
```

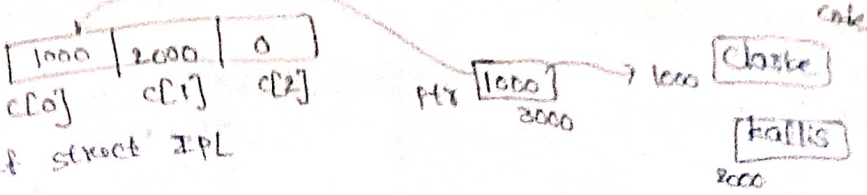
p	a	w	~~w~~ a	n	\0

23)
```
main()
{ int a=11;
  if (a==11)
  { int a=99;
  }
  else
    int a=55;
  Pf("%d", a); // 11.
```

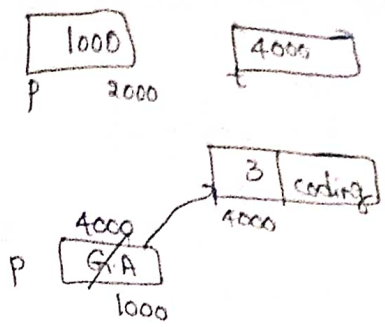
24) switch ON AC :-ANS)

25)
```
#define var 3
int main()
{
  char *cricket[var+0]={"clarke",
  char *ptr=cricket[1+no], "kallis"};
  Pf("%c", *++ptr); // e
  return 0;
}
```

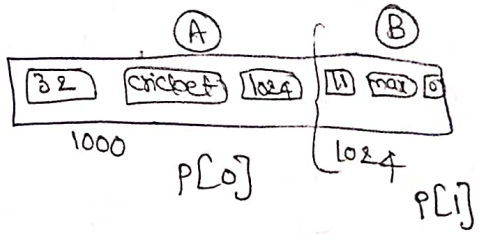
25:-)



26)
```
typedef struct IPL
{ int run;
  char name[11];
} csk;
void fun(csk **p)
{ csk *t;
  t = malloc(sizeof(csk));
  strcpy(t->name, "coding");
  t->run = 3;
  *p = t; → important.
}
void main()
{ csk *p;
  fun(&p);
  Pf("%s\n", p->name + p->run); //ing
}
```



27)
```
struct india
{ int RCB;
  char name[8];
  struct india *z;
};
void main()
{ struct india p[2] = {{32, "cricket", &p[1]}, {11, "maxwell", 0}};
  Pf("%d\n", p[0].z->RCB); // 11.
}
```



28) data structure is a c p. | → False

29) array is linear & dynamic data structure → False

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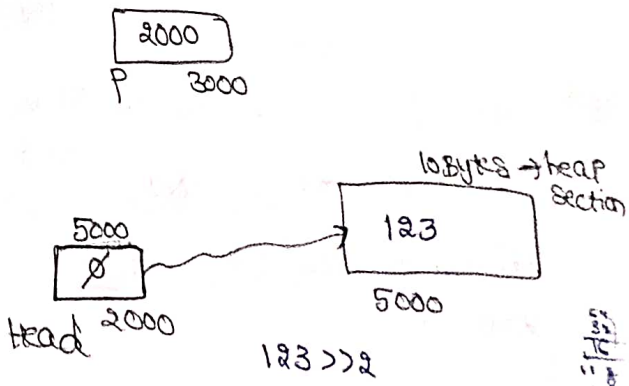
30)
```
#define S32 int
typedef struct st{
  S32 k;
  struct st *link;
}SLL;
// struct st *link
void add(SLL *p){
}
  *p = malloc(10);
int main(){
  SLL *head = NULL;
```



123 >> 2

000 | 1111 0 | 11
16 8 4 2 1

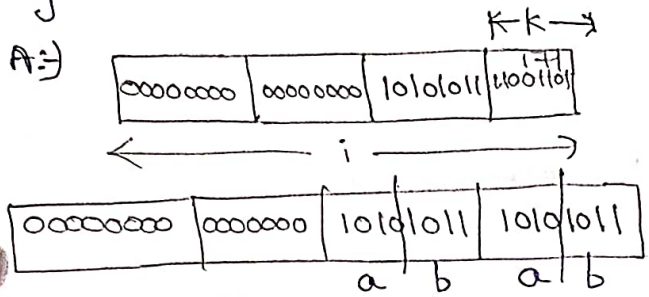
Ans:- 30.

16
8
4
2
30 →

```
add(&head);
head->k=128;
pf("%d\n", head->k>>2); //30
}
```

31)
```
union USA{
    int i;
    char k;
} V={0xabcd};
int main(){
union USA *IND = &V;
IND->k = 171;
Pf("%X\n", V.i);
return 0;
}
```

A:-)



Ans:- abab.

32)
```
int main(){
char *p[] = {"c", "ds", "project"};
Pf("%c\n", *(*(p+2)+2)+1);
return 0;
}
```

Ans:-

1000	1008	1016
2000	3000	4000
P[0]	P[1]	P[2]

A:-) P.

2000 "c"
3000 "ds"
4000 "project"

p[2][2]+1

'o'+1 → P.

33) pushpan sir
```
int a[] = {11, 22, 33, 44, 55, 66, 77};
short int (*p)[2] = &a;
Pf("%d\n", P[3][2]); //55.
return 0;
}
```

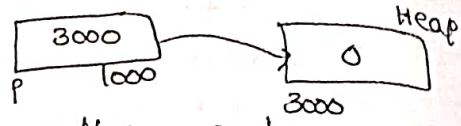
A:-)

11	22	33	44	55	66	77

P[0][0] P[0][1]
p[0] p[1] p[2][1]

34)
```
void *DmAC();
int main(){
    int *P = DmAC();
    *P = *P+3;
    pf("%d\n", *P); //3.
return 0;
}
void *DmAC(){
return calloc(1,4);
}
```

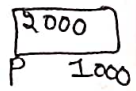
A:-)



take malloc :- Garbage

35)
```
char *P = "-3/4.5%12";
int main(){
double IPL = atof(P);
printf("%0.3lf\n", IPL);
return 0;
}
```

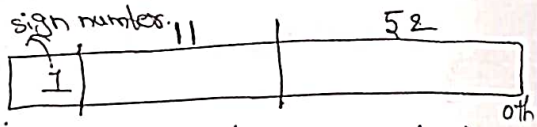
A:-) -3.000.



-3/4.5%12.
special.
⇒ IEEE 754

36)
```
double d=-99.345;
short int *p;
    p=(short int *)&d;
    p=p+3;
    Pf("%d\n", *p>>15&1);
return 0;
}
```

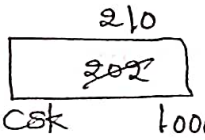
A:-)



A:-) 1. (negative number).

37)
```
int main(){
double *csk = 202;
++csk;
Pf("%d\n", 3+csk); // 234.
}
```

A:-)



3+csk → 210+3*8

= 234.

38)
```
int RCB =777;
const int *p = &RCB;
Pf("%d\n", *p+111); return 0;
}
```

A:) [777] RCB 1000

777 + 111 = 888.

P [1000] 2000

888.

39) int main(){
char s[1] = "5678";
char s1[2] = "pqrst";
char *PDF = s+8;
PDF = strcat(s1, s+2);
pf("%s\n", PDF+2);
return 0;
}

A:-) s [5 | 6 | 7 | 8 | 10] 1000

s1 [p | q | r | s | t | 7 | 8 | 10] 2000

PDF [1003] 3000

A:) rst78.

40) int a[] = {11, 22, 33, 44};
int *p = a+2;
pf("%d\n", ++*--p+2);
return 0;
}

A:-) [11 | 22 | 33 | 44] 23 --p

23+2 = 25.

41) void string (char *p){
p = p+2;
*p = '2';
}
int main(){
char *ptr;
ptr = "Sirji";
string (ptr);
pf("%s\n", ptr)
return 0;
}

A:-) error. ptr [G A] 2000 1000

P [2000] 2002 3000

Code section Sirji 2000

42) int x = 21;
void fun(int);
int main(){
fun(x);
pf("%d\n", x);
return 0;
}
void fun(int x1){
x = 45;
x1 = 78;
}

A:-) x [21] 2000 45 data

x1 [21] 3000 78

A:-) 45.

43) int a = 0x111;
short int b = 0111;
unsigned char csk = 11;
csk = csk+b+a;
pf("%d\n", csk);
return 0;
}

A:-) 0001 0001 (16 8 4 2 1) → 17

00 1001 001 (64 32 16 8 4 2 1) → 73

11+73+17 = 91.

64 8 1 73

73 17 0

Ans 101.

44) unsigned char k = 0;
if (k--)
pf("%d\n", k);
pf("%d\n", k); → -1 → 0 to 255
return 0;
}

A:-) 255.

45) short int i = 32;
if ((i&(i-1)) == 0) → power st
pf("power\n");
pf("star\n");
return 0;
}

A:-) star.